

Joshua Peek

The syntax analyzer, or parser, validates assignment statements by ensuring they conform to the defined grammar rules of the programming language. This process typically involves constructing a parse tree (or abstract syntax tree) based on the sequence of tokens provided by the lexical analyzer.

Grammar Section for Assignment Statements:

A common context-free grammar definition for an assignment statement might look like this:

```
<statement> ::= <identifier> = <expression> ;  
<expression> ::= <term> { ( + | - ) <term> }  
<term> ::= <factor> { ( * | / ) <factor> }  
<factor> ::= ( <expression> ) | <identifier> | <number>  
<identifier> ::= letter { letter | digit }  
<number> ::= digit { digit }
```

<https://www.geeksforgeeks.org/compiler-design/introduction-to-syntax-analysis-in-compiler-design/>

Aho, A. V., Ullman, J. D., Sethi, R., & Lam, M. S. (2007).
Compilers (2nd ed.). Pearson Education.